

When Surplus Capture Becomes “Exploitation”: A Contingent, Dynamic Account of Marxian Exploitation Under Modern Political Economy

In Marxian terms, *exploitation* is not merely the existence of profits or surplus capture, but a structural relation in which wage-labor systematically produces a surplus appropriated by owners or controllers of productive assets. Yet in modern political economy, measured profits can arise from many channels—competition-compatible returns, rents generated by market power, transfers, and risk premia—that are not isomorphic to the Marxian concept. We lay out a number of contingent factors that govern when surplus capture *from labor* becomes exploitation in the Marxian sense. We provide: (i) a clean conceptual decomposition separating labor-generated surplus from rents and transfers; (ii) sufficient conditions for Marxian exploitation as a class relation; (iii) a catalogue of contingent amplifiers (labor-market power, product-market power, state distortion, organizational control, financialization, and social reproduction); and (iv) a dynamic, finite-horizon selection framework in which time preference and trust determine which labor contracts and institutional “protocols” can persist without direct state control. The resulting view is compatible with Marx’s core structural insight while treating exploitation intensity as an equilibrium outcome of slow-moving social and institutional parameters.

Keywords: Marxian exploitation; surplus capture; labor wedge; monopsony; rent extraction; modern political economy; institutional trust; finite horizons.

1 Introduction: profits are not the concept

Classical political economy asked how stable social reproduction emerges from decentralized decisions [17, 20]. Marx accepts the ambition—a lawlike account of capitalism—but shifts the target from “natural prices” to the historically specific organization of labor under capital [7]. In Marx’s framework, profits are not simply a reward for abstinence, risk-bearing, or entrepreneurship. They express a *class relation* in which surplus labor is appropriated through the wage contract and property rights.

Modern political economy complicates the inference from profits to exploitation. Surplus capture can arise from markups, licensing, procurement rents, transfers, and asset revaluation [3, 18, 23]. The same accounting residual called “profit” can be driven by channels that do not primarily operate through the labor relation. This motivates a careful question:

*Under what conditions does surplus capture **from labor** become “exploitation” in Marxian terms, rather than merely rent extraction or competitive returns?*

Purpose, relevance, and why marry Marx to modern political economy. Our purpose in writing this paper is to give a *testable and institutionally grounded* account of when surplus capture *from labor* constitutes Marxian exploitation (a structural class relation), rather than a profit residual driven primarily by markups, political transfers, or risk premia. The relevance of linking Marxian exploitation to modern political economy is practical as well as conceptual: modern capitalism features labor-market frictions and wage-setting power [6, 9, 13], organizational control rights [1, 2, 24], and institutionalized rents [3, 23] that jointly determine whether surplus is generated and appropriated *through* the wage–labor relation. The “marriage” is therefore not a rebranding: it is a mapping from Marx’s class-structural concept to identifiable mechanisms—outside options, enforceability, and institutional closure—that govern the feasibility and persistence of labor wedges in equilibrium.

What we mean by “modern political economy”. By *modern political economy* we mean the contemporary, micro-founded study of how institutions and power shape economic outcomes: labor-market imperfections (search, matching, monopsony), contract and organization (control rights and monitoring), and political/institutional rent creation (licenses, procurement, and selective enforcement). We use these tools to *separate the concept* of exploitation (surplus labor appropriated via control) from the *channels* that amplify it (wage-setting power, rents, governance horizons, and state structure).

Research delta (new knowledge). Relative to classic debates that connect Marxian categories to competitive price theory [7, 8, 19, 21, 22], and relative to contemporary “extraction” accounts that often treat profits as the object of critique, we contribute three pieces of new, reusable structure: (i) a decomposition that separates labor-wedge surplus from rent/markup surplus (Proposition 1 and Lemma 1); (ii) a catalogue of institutionally interpretable wedge-enablers (monopsony, mobility constraints, organizational enforcement, governance horizons, and social reproduction); and (iii) a finite-horizon selection framework in which time preference and cognition determine which intertemporal labor “protocols” can persist without direct coercion (Sections 5.1–5.2).

We proceed in four moves. Section 2 states Marxian exploitation precisely and contrasts it with broader “extraction.” Section 3 provides an analytic decomposition separating labor-wedge surplus from rents. Section 4 lays out the contingent factors that jointly determine whether surplus capture is generated *by* labor under compulsion and *appropriated* by controllers. Section 5 gives a finite-horizon dynamic framework in which time preference and trust select which labor contracts and institutional arrangements can persist in a “natural” (low direct-control) order, drawing inspiration from a finite-horizon cutoff method and a potential/energy view of price formation.

2 Definitions: Marxian exploitation and its modern neighbors

2.1 Marxian exploitation as a structural class relation

Marx’s exploitation claim has two separable components:

1. **A production claim:** labor in production creates a surplus over and above what is required to reproduce labor-power.

2. **A distribution/appropriation claim:** the surplus is appropriated by those who own or control the means of production, through the wage-labor relation and property rights.

Definition 1 (Marxian exploitation (technical sense)). Consider a capitalist workplace where (i) workers sell labor-power for a wage; (ii) the employer controls production and owns output; and (iii) the working day contains *necessary labor time* (reproducing the wage) and *surplus labor time* (producing value beyond the wage). Marxian exploitation obtains when surplus labor time is positive and the corresponding surplus is systematically appropriated by the employer class [7].

This definition does not require fraud or a violated contract; it treats exploitation as compatible with legally “free” wage exchange.

2.2 Extraction: a broader umbrella

Many contemporary accounts use *extraction* to denote durable surplus capture enabled by power, barriers, or institutional privilege. This includes, but is not limited to, Marxian exploitation: rents from monopoly or licensing [3], transfers sustained by political power [18], or surplus capture by controllers of large organizations [5]. Marxian exploitation is a *subset*: extraction *from labor* under wage-labor and control of production.

2.3 Why “profit” is an unreliable indicator

Profit is an accounting residual. It may be:

- a competitive return to capital/services of entrepreneurship;
- a rent generated by product-market power or state-created scarcity;
- a transfer from taxpayers/consumers;
- a risk premium;
- or a surplus generated by a labor wedge (underpayment relative to contribution).

Therefore the mapping “profit $\Rightarrow$ exploitation” is not an identity. The task is to isolate when surplus capture is specifically a labor-appropriation phenomenon.

3 Decomposition: separating labor-wedge surplus from rents

3.1 A minimal surplus accounting

Let value added be Y . Let total labor compensation be W . Let measured operating surplus be Π (broadly, profits before interest/taxes in national accounts, plus mixed income issues). The aggregate identity is:

$$Y = W + \Pi. \tag{1}$$

Marxian exploitation concerns the mechanism by which part of Y produced by labor is not returned as W but captured as surplus by controllers.

3.2 The labor-wedge concept

A modern, non-labor-theory-of-value operationalization is the wedge between what labor contributes at the margin and what it is paid:

$$\text{Labor wedge} \equiv MRPL - w, \quad (2)$$

where $MRPL$ is marginal revenue product of labor and w the wage. If $MRPL > w$ persistently, labor is underpaid relative to contribution, generating a wedge that can be interpreted as a channel through which surplus capture *from labor* occurs.

3.3 Rents can generate profits without labor exploitation

Proposition 1 (Rents do not imply a labor wedge). *There exist economies with strictly positive measured profits $\Pi > 0$ in which the labor wedge is identically zero, i.e.*

$$w = MRPL$$

at the firm optimum. Hence positive profits do not logically imply surplus capture from labor via underpayment.

Proof. We give an explicit construction.

Environment. Consider a firm with a strictly increasing, strictly concave production function $q = F(L)$, $F'(L) > 0$, $F''(L) < 0$. The firm sells output facing an inverse demand curve $p = P(q)$ with $P(q) > 0$ and $P'(q) < 0$ for $q > 0$. Let revenue be

$$R(q) := P(q) q.$$

Assume R is differentiable and strictly concave in q on the relevant range (this holds for standard constant-elasticity demand, among others). The firm hires labor at wage $w > 0$ in a competitive labor market, taking w as given, and chooses $L \geq 0$ to maximize

$$\pi(L) := R(F(L)) - wL.$$

First-order condition and “no wedge.” Because $R \circ F$ is strictly concave in L (composition of concave functions with F increasing), π is strictly concave in L , so any interior optimum is unique and characterized by the first-order condition:

$$\frac{d}{dL} \pi(L) = R'(F(L)) F'(L) - w = 0.$$

Define marginal revenue product of labor as

$$MRPL(L) := \frac{d}{dL} (R(F(L))) = R'(F(L)) F'(L).$$

Thus at the optimizer L^* we have

$$w = MRPL(L^*),$$

so the labor wedge $MRPL - w$ is exactly zero at the profit-maximizing choice.

Yet profits can be strictly positive. Pick any case with downward-sloping demand so the firm has product-market power. Then the revenue-maximizing price exceeds marginal cost in the standard monopoly logic. Concretely, write the marginal revenue as $MR(q) = R'(q)$ and the marginal physical product as $MP(L) = F'(L)$. At the optimum, $w = MR(F(L^*)) MP(L^*)$.

Now compare this to the competitive benchmark price-taking case where output price is constant $p = \bar{p}$. Under monopoly, for the same L , we have $MR(q) < p(q)$ whenever demand is downward sloping (since $MR(q) = p(q) + qp'(q)$ and $p'(q) < 0$). The wedge relevant for *product-market power* is between price and marginal revenue, not between $MRPL$ and w .

To see strict positivity directly, take a constant-elasticity demand curve

$$P(q) = q^{-1/\varepsilon} \quad \text{with} \quad \varepsilon > 1,$$

so $R(q) = q^{1-1/\varepsilon}$ and $R'(q) = (1 - 1/\varepsilon) q^{-1/\varepsilon}$. Let $F(L) = L^\alpha$ for $\alpha \in (0, 1)$. Then

$$\pi(L) = (L^\alpha)^{1-1/\varepsilon} - wL = L^{\alpha(1-1/\varepsilon)} - wL.$$

Since $\alpha(1 - 1/\varepsilon) \in (0, 1)$, the first term is strictly concave and dominates near $L = 0$, so the maximizer $L^* > 0$ exists. At that L^* , profits satisfy

$$\pi(L^*) = R(F(L^*)) - wL^*.$$

Using the first-order condition,

$$w = R'(F(L^*))F'(L^*),$$

strict concavity of $R \circ F$ implies the tangent line at L^* lies above the function elsewhere; in particular

$$R(F(L^*)) > R'(F(L^*))F'(L^*)L^* = wL^*$$

whenever $R \circ F$ is strictly concave and $L^* > 0$.¹ Hence $\pi(L^*) > 0$.

Interpretation. Profits here arise from product-market power (a markup/rent channel), while the firm still sets labor so that $w = MRPL$. Therefore positive profits do not imply a labor wedge, and thus do not logically imply Marxian exploitation understood as systematic surplus capture *from labor via underpayment*. $\square$

Remark 1 (What this does and does not show). Proposition 1 does *not* say that monopoly profits are normatively innocent, nor that they cannot coexist with labor exploitation. It only establishes a logical non-implication: profits can be large even when the labor wedge is zero.

3.4 A simple decomposition: rent-driven vs wedge-driven surplus

The prior proposition shows that Π can be large even when $MRPL - w = 0$. To connect this to measurement, it is useful to isolate a wedge-driven component of operating surplus from a rent/markup component.

Lemma 1 (Accounting decomposition at the firm level). *Consider any firm with revenue $R(L)$ and wage bill wL , where R is differentiable and concave in L . For any reference wage*

¹This is the strict inequality $g(x) > g'(x)x$ for strictly concave g with $g(0) = 0$ and $x > 0$; apply $g(L) = R(F(L))$.

$\tilde{w}$, define the wedge component at employment L by

$$\Pi^{\text{wedge}}(L; \tilde{w}) := \int_0^L (MRPL(\ell) - \tilde{w}) d\ell, \quad MRPL(\ell) := R'(\ell).$$

Then profits satisfy

$$\pi(L) = R(L) - wL = \underbrace{\Pi^{\text{wedge}}(L; w)}_{\text{wedge-driven}}.$$

If instead one evaluates surplus relative to a competitive (or counterfactual) wage $\tilde{w}$ (e.g. a benchmark from another regime), then

$$\pi(L) = \Pi^{\text{wedge}}(L; \tilde{w}) + (\tilde{w} - w)L.$$

Proof. The first identity is just the Fundamental Theorem of Calculus: since $R(0) = 0$ (normalize) and R is differentiable,

$$R(L) = \int_0^L R'(\ell) d\ell = \int_0^L MRPL(\ell) d\ell,$$

so

$$\pi(L) = \int_0^L MRPL(\ell) d\ell - wL = \int_0^L (MRPL(\ell) - w) d\ell.$$

For the second, add and subtract $\tilde{w}$ inside the integral and simplify:

$$\int_0^L (MRPL(\ell) - w) d\ell = \int_0^L (MRPL(\ell) - \tilde{w}) d\ell + \int_0^L (\tilde{w} - w) d\ell = \Pi^{\text{wedge}}(L; \tilde{w}) + (\tilde{w} - w)L.$$

□

Remark 2 (Why this helps empirically). The lemma clarifies what must be pinned down to call surplus “wedge-driven”: a benchmark wage $\tilde{w}$ (competitive, counterfactual, or a structural outside-option wage). Without such a benchmark, the statement “profits came from labor underpayment” is not identified from accounting alone.

3.5 Monopsony and bargaining structurally sustain the labor wage

Under imperfect labor-market competition, firms can have wage-setting power. In canonical monopsony models, the wage markdown depends on the elasticity of labor supply to the firm [6]. Bargaining models similarly make wages a function of outside options and bargaining weights, not purely marginal products [9, 13]. These are modern microfoundations for the persistence of a labor wedge.

Proposition 2 (A canonical sufficient condition for a positive wage markdown). *Suppose a firm faces an upward-sloping labor supply $L(w)$ with elasticity*

$$\varepsilon(w) := \frac{w}{L(w)} L'(w) \in (0, \infty).$$

If the firm chooses w to maximize profits $\pi(w) = R(L(w)) - wL(w)$ with $R'(\cdot) > 0$, then at any interior optimum w^ the wage satisfies*

$$w^* = \frac{\varepsilon(w^*)}{1 + \varepsilon(w^*)} MRPL^*,$$

where $MRPL^* := \frac{d}{dL} R(L)|_{L=L(w^*)}$. In particular, $w^* < MRPL^*$ whenever $\varepsilon(w^*) < \infty$.

Proof. Differentiate $\pi(w)$:

$$\pi'(w) = R'(L(w))L'(w) - L(w) - wL'(w).$$

At an interior optimum, $\pi'(w^*) = 0$, so

$$R'(L(w^*))L'(w^*) = L(w^*) + w^*L'(w^*).$$

Divide both sides by $L'(w^*) > 0$:

$$R'(L(w^*)) = \frac{L(w^*)}{L'(w^*)} + w^*.$$

Rearrange:

$$w^* = R'(L(w^*)) - \frac{L(w^*)}{L'(w^*)}.$$

Using $\varepsilon(w^*) = \frac{w^*}{L(w^*)}L'(w^*)$ gives $\frac{L(w^*)}{L'(w^*)} = \frac{w^*}{\varepsilon(w^*)}$, so

$$w^* = MRPL^* - \frac{w^*}{\varepsilon(w^*)} \implies w^* \left(1 + \frac{1}{\varepsilon(w^*)}\right) = MRPL^*,$$

hence

$$w^* = \frac{\varepsilon(w^*)}{1 + \varepsilon(w^*)} MRPL^* < MRPL^*.$$

□

4 The zoo of contingent factors

This section catalogues the major factor families that govern whether surplus capture becomes Marxian exploitation (surplus labor appropriated through wage-labor and control) rather than merely rent extraction or competitive return. The organizing principle is: exploitation intensity rises when (i) *the labor wedge is feasible and enforceable* and (ii) *the resulting surplus is appropriated by controllers as a stable class outcome*.

4.1 Structural prerequisites (non-contingent core)

These are the baseline conditions without which Marxian exploitation (as Marx defines it) is not well-posed:

1. **Separation from means of production:** workers lack independent access to productive assets and must sell labor-power [7].
2. **Wage-labor contract:** labor-power is purchased for a wage; employer directs labor and owns output.
3. **Residual claimancy and control rights:** owners/controllers claim the residual surplus and command production decisions [1, 2, 24].
4. **Competitive compulsion for capital:** firms must generate surplus to survive competition and finance accumulation [7].

These conditions make surplus appropriation a systemic feature rather than an episodic abuse.

4.2 Labor-market power and outside options (wedge feasibility)

The labor wedge $MRPL - w$ becomes feasible when workers' exit is costly or blocked. The contingent determinants include:

1. **Monopsony/oligopsony:** labor supply to the firm is inelastic due to concentration, geography, or search frictions [6, 9].
2. **Search frictions and matching:** unemployment risk and search costs weaken outside options and bargaining position [13].
3. **Mobility constraints:** housing, transport, family obligations, immigration/visa ties, occupational licensing.
4. **Contractual restraints:** noncompetes, training repayment agreements, blacklists; these reduce effective labor supply elasticity.
5. **Segmentation and discrimination:** segmented submarkets prevent arbitrage to higher wages, sustaining wedges.
6. **Debt and liquidity constraints:** inability to smooth consumption increases acceptance of low wages (a time-preference amplifier; see Section 5).

4.3 Product-market power and rents (raising the prize)

Product-market power does not automatically create labor exploitation (Proposition 1), but it raises the value of control and can amplify wedge extraction:

1. **Markups and barriers to entry:** monopoly, cartelization, IP barriers, network effects.
2. **State-created scarcity:** licenses, quotas, procurement gatekeeping [3].
3. **Trade protection and local monopolies:** tariffs, exclusive franchises [23].

Mechanism: higher rents raise the return to investments in control (monitoring, union suppression, exit barriers), making labor exploitation *more profitable* even if not logically implied.

4.4 State distortion and bureaucratic choke points

State structure can either discipline rents or create them. When the state is a rent allocator, surplus capture tends to become predatory and can couple tightly to labor exploitation:

1. **Administrative allocation:** credit, land, and permits are mediated by officials, creating office-based rents [11].
2. **Selective enforcement:** legal uncertainty and discretionary inspection create compliance rents.
3. **Clientelism:** labor allocation and welfare access tied to political brokers weakens worker exit.

In such environments, the same apparatus that creates rents can also constrain labor mobility, strengthening the labor wedge.

4.5 Organization, monitoring technology, and “middle management”

Middle management is not inherently extractive. It becomes a predation complement when the profit strategy relies on enforcing a wedge or intensifying labor:

1. **Team production monitoring:** supervision solves internal shirking problems but can also enforce speedups and unpaid intensity [1].
2. **Information asymmetries:** control over metrics and evaluation can suppress wage growth and weaken bargaining.
3. **Surveillance and algorithmic management:** reduces monitoring cost, making wedge enforcement cheaper.
4. **Credential ladders and elite closure:** management becomes a gatekeeping layer when advancement is controlled by bureaucratic rules rather than open contestation.

In “open” district economies, management can be a permeable rung in a broad local elite; in closed bureaucracies it becomes a distinct enforcement layer that stabilizes extraction (an “outer party” mechanism).

4.6 Corporate governance and financialization

When payout and valuation regimes reward short-run cash extraction, firms are incentivized to increase surplus without building capability:

1. **Stock-based pay:** shifts objectives toward measurable short-run financial metrics [5].
2. **Buybacks/dividends:** increase pressure to cut labor costs and underinvest in training [4].
3. **Private equity and leveraged governance:** debt service imposes near-term extraction constraints.

This family often interacts strongly with time preference and finite-horizon decision making (Section 5).

4.7 Social reproduction and “reproduction costs” of labor-power

In Marxian terms, the wage is linked to the reproduction of labor-power. Institutions that privatize or socialize reproduction costs (healthcare, childcare, housing) shift the feasible wage floor:

1. **Public provisioning:** can raise worker outside options and reduce acceptance of extractive contracts.
2. **Household burden shifting:** can allow low wages to persist by offloading costs to unpaid household labor.
3. **Migration and remittances:** can subsidize low wages in receiving regions while externalizing reproduction to sending regions.

4.8 Trust, civic density, and informal enforcement

High generalized trust and dense associations can substitute for bureaucratic enforcement and reduce the profitability of predation:

1. **Horizontal sanctioning:** reputational penalties deter opportunism in repeated relations.
2. **Collective voice:** unions, cooperatives, associations can raise wages and reduce wedges.
3. **Institutional performance:** civic traditions can improve state quality and reduce rent allocation [15].

Industrial district research emphasizes how repeated inter-firm relations and local institutions support cooperation and diffusion, potentially limiting predatory extraction [12, 16].

5 Dynamics with finite horizons: time preference, cognition, protocol selection, and market thickness

If exploitation intensity is contingent, the key dynamic question is: *which environments select for extractive labor relations, and which select against them, absent direct state control?* This section supplies two technical primitives used repeatedly:

1. a finite-horizon cutoff theorem for acceptance of “front-loaded sweetener / back-loaded burn” arrangements (including a symmetric producer-side cutoff for wage investment); and
2. a potential/“energy” formulation of wage/price clearing based on cumulative supply–demand imbalance, augmented by a measurable trust-mediated participation layer that feeds local thickness, together with deterministic and stochastic (finite- n , finite-“temperature”) dynamics. Both are written in a form designed for finite T and finite- n reasoning.

5.1 Finite-horizon evaluation and a cutoff in time preference

Setup (one-shot Bellman reduction). Fix a finite horizon $T \in \mathbb{N}$. Let an agent evaluate a labor contract (or institutional protocol) by a stream of per-period increments $\{\Delta_t\}_{t=1}^T$ relative to an outside option:

$$\Delta_t := U_t(P) - U_t(S), \quad t = 1, \dots, T.$$

With discount factor $\delta \in (0, 1)$, the discounted increment is

$$\Delta U(\delta) = \sum_{t=1}^T \delta^{t-1} \Delta_t. \quad (3)$$

The one-shot decision rule is: choose P iff $\Delta U(\delta) \geq 0$. A key empirical regularity of extractive arrangements is *front-loaded sweeteners with back-loaded costs* (signing bonuses, short-run relief, or option value now; lock-in, wage stagnation, intensity creep later).

Proposition 3 (Single-crossing horizon profile). *There exists an index $m \in \{1, \dots, T-1\}$ such that*

$$\Delta_t > 0 \text{ for } t \leq m, \quad \Delta_t < 0 \text{ for } t > m,$$

and additionally $\Delta_1 > 0$ and $\Delta_T < 0$. Assume also that

$$\Delta U(1) = \sum_{t=1}^T \Delta_t < 0.$$

Lemma 2 (Descartes one-sign-change bound). *Let $P(x) = \sum_{k=0}^n a_k x^k$ be a real polynomial. If the nonzero coefficients (a_k) exhibit exactly one sign change when read in increasing k , then P has at most one positive real root.*

Proof. This is the one-sign-change case of Descartes' Rule of Signs. $\square$

Theorem 1 (Finite-horizon cutoff in time preference). *Assume 3. Then there exists a unique cutoff $\delta^* \in (0, 1)$ such that*

$$\Delta U(\delta) \geq 0 \iff \delta \leq \delta^*.$$

Equivalently: the alternative protocol P is chosen iff the agent is sufficiently impatient (high time preference, low δ).

Proof. Define the polynomial $P(x) := \sum_{t=1}^T \Delta_t x^{t-1}$, so $\Delta U(\delta) = P(\delta)$ for $\delta \in (0, 1)$. By 3, the coefficient sequence (Δ_t) has exactly one sign change $+\rightarrow -$, hence P has at most one positive root by 2. Moreover $P(0) = \Delta_1 > 0$ and $P(1) = \sum_{t=1}^T \Delta_t < 0$, so continuity implies at least one root in $(0, 1)$. Therefore there is exactly one root $\delta^* \in (0, 1)$, and the sign pattern yields $P(\delta) \geq 0 \Leftrightarrow \delta \leq \delta^*$. $\square$

Remark 3 (Scope and limits of the single-crossing payoff profile). The single-crossing structure in Proposition 3 is a *sufficient* condition that delivers a sharp cutoff in δ . It fits a broad family of “front-loaded sweetener / back-loaded burn” arrangements (signing bonuses with lock-in, delayed vesting with forfeiture risk, or early wage premia with later intensity creep). It is not claimed to exhaust all exploitative relations: multi-stage contracts can generate multiple sign changes, and some institutional regimes produce persistent negative increments rather than a single reversal. In those cases the acceptance set may be non-monotone in δ , and the cutoff result becomes a bracketing device rather than an exact characterization. The empirical role of the theorem is therefore diagnostic: when observed contract profiles are approximately single-crossing, impatience alone predicts systematic selection into dominated arrangements.

Remark 4 (Selection interpretation). Theorem 1 isolates a purely *dynamic selection* channel: if an arrangement delivers early private gains and later diffuse losses (a single sign change in Δ_t), then high-time-preference types self-select into it even when the long-run total is negative. This is a formal way to say “impatience makes people tolerate extraction”: the mechanism is discounting under finite T , not a moral claim.

5.2 Cognition and probability distortion

The finite-horizon cutoff in Section 5.1 shows how *time preference* alone can select agents into front-loaded/back-loaded-negative arrangements. But many real-world “sweetener/burn” protocols are not purely deterministic. They embed *tail risks* (termination triggers, hours volatility, contingent bonuses, delayed vesting with forfeiture, health or injury exposure),

and they advertise *lottery-like upsides* (promotion narratives, performance prizes, option value) whose realized probabilities are hard to learn ex ante. In such environments, selection is driven not only by δ but also by systematic *misperception of risk*. A convenient reduced-form representation is to let cognition g parameterize how close subjective probability assessment is to objective frequency. Low- g types may overweight small-probability upside and underweight small-probability downside, raising the perceived early increments and muting later losses. This effectively *reshapes the horizon profile* $\{\Delta_t\}$ into a cognition-dependent profile $\{\tilde{\Delta}_t(g)\}$, shifting the acceptance cutoff $\delta^*(g)$ and thereby changing which protocols can persist in equilibrium. Hence “cognition” is not a behavioral flourish: it is a second, conceptually distinct selection margin that can sustain extractive protocols even when fully informed, patient agents would reject them.

Many extractive arrangements are risky in ways that are easy to mis-evaluate: layoffs, variable hours, promotion lotteries, bonus clawbacks, performance-based terminations, or health shocks. If agents distort probabilities, then the deterministic increment Δ_t in Section 5.1 should be replaced by a *certainty-equivalent* increment $\tilde{\Delta}_t(g)$ that depends on a cognition parameter g governing the strength (and possibly asymmetry) of distortion.

Risky increment as a lottery. Fix a finite horizon T . At each date t , let the protocol-vs-outside-option payoff difference be a random variable X_t (measurable w.r.t. the date- t information set), interpreted as the *per-period increment* under the protocol:

$$X_t := U_t(P) - U_t(S).$$

For analytic transparency, we use a three-outcome representation that captures many labor-market risks. Let E_t^+ be a “good” event (bonus/promotion) and E_t^- a “bad” event (layoff/penalty), and assume the events are disjoint: $E_t^+ \cap E_t^- = \emptyset$. Let $\mathbb{P}(E_t^+) = p_t^+ \in (0, 1)$ and $\mathbb{P}(E_t^-) = p_t^- \in (0, 1)$, with $p_t^+ + p_t^- < 1$. Then

$$X_t = \begin{cases} x_t + G_t, & \text{with prob. } p_t^+, \\ x_t - L_t, & \text{with prob. } p_t^-, \\ x_t, & \text{with prob. } 1 - p_t^+ - p_t^-, \end{cases} \quad (4)$$

where $x_t \in \mathbb{R}$ is a sure component (for example, a baseline wage difference), $G_t \geq 0$ is an upside magnitude, and $L_t \geq 0$ a downside magnitude. This form covers promotion lotteries, performance termination risk, hours volatility, and health/expense shocks under the protocol.

CPT-style certainty equivalent with Prelec weighting. Let $u : \mathbb{R} \rightarrow \mathbb{R}$ be a strictly increasing, continuous utility index with $u(0) = 0$ and u concave near 0. We define the date- t *subjective* certainty equivalent of X_t as

$$\tilde{\Delta}_t(g) := u(x_t) + w^+(p_t^+; g) u(G_t) - w^-(p_t^-; g) u(L_t), \quad (5)$$

where w^+ and w^- are probability-weighting functions for gains and losses, respectively. A canonical choice is Prelec’s form [14]:

$$w^\pm(p; g) = \exp\left(-\beta_\pm(g) (-\ln p)^{\alpha_\pm(g)}\right), \quad p \in (0, 1), \quad (6)$$

with parameters $\alpha_\pm(g) \in (0, 1]$ and $\beta_\pm(g) > 0$. When $\alpha_\pm = 1$ and $\beta_\pm = 1$, $w^\pm(p) = p$ (no distortion). For $\alpha_\pm < 1$, the weighting function exhibits the familiar inverse-S shape

(overweighting small probabilities and underweighting moderate ones).

A standard CPT specification uses a value function v around a reference point 0,

$$v(z) = \begin{cases} z^{\eta_+}, & z \geq 0, \\ -\lambda(-z)^{\eta_-}, & z < 0, \end{cases} \quad \eta_{\pm} \in (0, 1], \lambda \geq 1,$$

and replaces $u(\cdot)$ in (5) by $v(\cdot)$. None of the comparative-statics results below change as long as v is increasing and $v(G_t), v(L_t) \geq 0$ for $G_t, L_t \geq 0$.

Cognition as reduced distortion. We model cognition (or “ g factor”) as shifting the weighting functions toward objective probabilities.

Assumption 1 (Distortion shrinks with cognition). For each sign $\pm$ and each $p \in (0, 1)$, $w^{\pm}(p; g)$ is continuous in g and

$$\lim_{g \rightarrow \infty} w^{\pm}(p; g) = p.$$

Moreover, on the probability ranges relevant for extractive protocols:

1. (Lottery optimism shrinks) for the small-probability gain range of interest, $\partial_g w^+(p; g) \leq 0$;
2. (Downside neglect shrinks) for the small-probability loss range of interest, $\partial_g w^-(p; g) \geq 0$.

Assumption 1 allows for the empirically plausible asymmetry that low- g agents may overweight upside lotteries (e.g. promotion chances) while underweighting downside risks (e.g. termination hazards). It can be implemented in (6) by taking $\alpha_{\pm}(g) \uparrow 1$ and $\beta_{\pm}(g) \rightarrow 1$ as g rises, together with parameter restrictions that yield the stated signs on the relevant p ranges.

Dynamic acceptance with distorted probabilities. Replace Δ_t in (3) by $\tilde{\Delta}_t(g)$ and define

$$\widetilde{\Delta U}(\delta, g) := \sum_{t=1}^T \delta^{t-1} \tilde{\Delta}_t(g). \quad (7)$$

The (behavioral) decision rule is: choose protocol P iff $\widetilde{\Delta U}(\delta, g) \geq 0$.

Single-crossing and the behavioral cutoff. To retain the crisp finite-horizon cutoff logic, assume the certainty-equivalent increment sequence is front-loaded-positive/back-loaded-negative.

Assumption 2 (Behavioral single-crossing profile). For each fixed g , there exists $m \in \{1, \dots, T-1\}$ such that

$$\tilde{\Delta}_t(g) > 0 \text{ for } t \leq m, \quad \tilde{\Delta}_t(g) < 0 \text{ for } t > m,$$

and $\sum_{t=1}^T \tilde{\Delta}_t(g) < 0$.

Under Assumption 2, the same Descartes-sign-change argument as in Theorem 1 implies the existence of a *unique* behavioral cutoff $\delta^*(g) \in (0, 1)$ such that

$$\widetilde{\Delta U}(\delta, g) \geq 0 \iff \delta \leq \delta^*(g).$$

Thus, for each cognition level g , the protocol is accepted exactly by sufficiently impatient types.

Comparative statics: higher cognition reduces acceptance of extractive risk profiles.

The key content is how $\delta^*(g)$ moves with g .

Lemma 3 (Sufficient condition for $\partial_g \tilde{\Delta}_t(g) \leq 0$). *Assume $G_t, L_t \geq 0$ and $u(\cdot) \geq 0$ on $\mathbb{R}_+$. If on the relevant probability ranges we have $\partial_g w^+(p_t^+; g) \leq 0$ and $\partial_g w^-(p_t^-; g) \geq 0$, then*

$$\partial_g \tilde{\Delta}_t(g) \leq 0.$$

Proof. Differentiate (5):

$$\partial_g \tilde{\Delta}_t(g) = \partial_g w^+(p_t^+; g) u(G_t) - \partial_g w^-(p_t^-; g) u(L_t) \leq 0,$$

since $u(G_t), u(L_t) \geq 0$ and the derivatives have opposite signs. $\square$

Proposition 4 (Cognition shrinks the acceptance region). *Suppose Assumptions 1 and 2 hold, and assume the objective risks satisfy $G_t, L_t \geq 0$ with $u(\cdot) \geq 0$ on $\mathbb{R}_+$. If, in addition, the sign conditions of Lemma 3 hold for all t , then the unique cutoff $\delta^*(g)$ is nonincreasing in g . In particular, higher cognition weakly reduces the set of discount factors for which the agent accepts the protocol.*

Proof. Fix g and define the polynomial in δ :

$$F(\delta, g) := \widetilde{\Delta U}(\delta, g) = \sum_{t=1}^T \delta^{t-1} \tilde{\Delta}_t(g).$$

By Assumption 2 and the same argument as Theorem 1, there is a unique $\delta^*(g) \in (0, 1)$ such that $F(\delta, g) \geq 0$ iff $\delta \leq \delta^*(g)$. Hence $F(\delta, g)$ is positive for $\delta < \delta^*(g)$ and negative for $\delta > \delta^*(g)$, so F crosses from $+$ to $-$ at $\delta^*(g)$. Moreover, since the coefficient sequence $(\tilde{\Delta}_t(g))_{t=1}^T$ has exactly one sign change, Descartes' Rule of Signs implies $F(\cdot, g)$ has at most one positive real root counted *with multiplicity*. Because $\delta^*(g) \in (0, 1)$ is a positive root, it must be simple, hence $\partial_\delta F(\delta^*(g), g) \neq 0$. The fact that F changes sign from $+$ to $-$ then forces $\partial_\delta F(\delta^*(g), g) < 0$.

Now differentiate the identity $F(\delta^*(g), g) = 0$ using the implicit-function theorem:

$$\delta^{*'}(g) = -\frac{\partial_g F(\delta^*(g), g)}{\partial_\delta F(\delta^*(g), g)}.$$

By Lemma 3, for each t we have $\partial_g \tilde{\Delta}_t(g) \leq 0$. Therefore, for any $\delta \in (0, 1)$,

$$\partial_g F(\delta, g) = \sum_{t=1}^T \delta^{t-1} \partial_g \tilde{\Delta}_t(g) \leq 0.$$

Thus $\partial_g F(\delta^*(g), g) \leq 0$ while $\partial_\delta F(\delta^*(g), g) < 0$, yielding

$$\frac{d}{dg} \delta^*(g) \leq 0.$$

$\square$

Remark 5 (When does $\partial_g \tilde{\Delta}_t(g) \leq 0$ hold?). From (5), we have

$$\partial_g \tilde{\Delta}_t(g) = \partial_g w^+(p_t^+; g) u(G_t) - \partial_g w^-(p_t^-; g) u(L_t).$$

Under Assumption 1, $\partial_g w^+ \leq 0$ on the relevant small- p gain range and $\partial_g w^- \geq 0$ on the relevant small- p loss range, so $\partial_g \tilde{\Delta}_t(g) \leq 0$ holds whenever $u(G_t), u(L_t) \geq 0$. Intuitively: increasing cognition (i) deflates overweighted upside lotteries and (ii) inflates neglected downside risks, reducing the perceived net increment.

Remark 6 (Interpretation for “extractive protocol selection”). Combine Proposition 4 with Theorem 1: cognition operates like a *selection shifter*. For a fixed horizon profile (front-loaded sweetener, back-loaded downside), higher g lowers $\delta^*(g)$, shrinking the population of types who accept dominated risky protocols. Thus a society/firms facing lower average cognition (or systematically more distorted beliefs) can sustain more extraction by horizon engineering even with identical objective payoffs.

Institutional sources of δ and g . The parameters δ (effective time preference) and g (risk cognition) are not treated as purely psychological constants. They are shaped by institutional context and social reproduction: liquidity constraints, debt, housing insecurity, and health-cost exposure compress horizons (lower effective δ), while education, transparent contracting, unions/collective voice, and enforcement of disclosure norms raise g by improving learning and reducing misperceived tail risks. Conversely, environments that externalize reproduction costs onto households, or that rely on contingent hours/bonuses and opaque evaluation, can systematically lower g and make horizon-engineered protocols more stable in equilibrium.

5.3 Producer impatience and wage refusal as a symmetric cutoff

Let a firm consider a wage-raising policy that changes profit by $\Delta\pi_t$ (costs now, benefits later through retention/quality). With firm discount factor δ_F , the net present gain is

$$\Delta\Pi(\delta_F) = \sum_{t=1}^T \delta_F^{t-1} \Delta\pi_t.$$

If $\Delta\pi_t < 0$ early and $\Delta\pi_t > 0$ later (wage investment is back-loaded), and $\sum_{t=1}^T \Delta\pi_t > 0$ (patiently evaluated, it is beneficial), then the same polynomial argument yields a unique cutoff $\delta_F^* \in (0, 1)$ such that only sufficiently patient firms raise wages. Thus high time preference producers systematically resist wage increases whenever the gains from wage investment are back-loaded.

5.4 Market thickness as an “energy” barrier to predation

We treat the wage w as a clearing “price” between (i) a distribution of wage offers by firms (bids) and (ii) a distribution of reservation wages by workers (asks). The representation is a *continuum clearing* framework whose primitives have direct interpretations: density near clearing and sensitivity of clearing wages to small deformations of the environment.

Assumption 3 (Regularity of bid/ask densities). Fix a time t . The bid and ask depth densities $b(\cdot, t)$ and $a(\cdot, t)$ satisfy:

1. $a(\cdot, t), b(\cdot, t) \in L^1(\mathbb{R}) \cap C(\mathbb{R})$ and are nonnegative.

2. Total masses are finite and strictly positive:

$$A(t) := \int_{\mathbb{R}} a(w, t) dw \in (0, \infty), \quad B(t) := \int_{\mathbb{R}} b(w, t) dw \in (0, \infty).$$

3. (Local boundedness) For any compact interval $K \subset \mathbb{R}$, $\sup_{w \in K} a(w, t) < \infty$ and $\sup_{w \in K} b(w, t) < \infty$.

When we study dynamics in t , we additionally assume $(w, t) \mapsto a(w, t), b(w, t)$ are continuous and admit the derivatives stated where needed.

Methodological note (interpretation of “bid/ask” and stochastic adjustment). The bid and ask objects here are *reduced-form distributions* of wage offers (firms) and reservation wages (workers) at a given time. They define a transparent “excess demand” function $D(w, t)$ whose slope $\rho(w, t)$ measures local market thickness (how much mass lies near the clearing wage), yielding simple sensitivity bounds. The Langevin specification in (10) is a diffusion approximation to finite participation and noisy sampling of offers/reservations (finite n). It provides a compact way to state how thin markets (low κ) mechanically amplify the wage effect of small institutional deformations.

Cumulatives and imbalance. Define

$$B^{\leq}(w, t) := \int_{-\infty}^w b(y, t) dy, \quad S^{\geq}(w, t) := \int_w^{\infty} a(y, t) dy,$$

and the cumulative imbalance (excess demand)

$$D(w, t) := B^{\leq}(w, t) - S^{\geq}(w, t).$$

A clearing wage $w^*(t)$ is any solution of $D(w^*(t), t) = 0$.

Lemma 4 (Regularity, monotonicity, and derivative identity). *Under Assumption 3, for each fixed t :*

1. $B^{\leq}(\cdot, t)$ and $S^{\geq}(\cdot, t)$ are absolutely continuous and continuous on $\mathbb{R}$.
2. $D(\cdot, t)$ is continuous and nondecreasing.
3. For a.e. $w \in \mathbb{R}$,

$$\partial_w D(w, t) = a(w, t) + b(w, t) =: \rho(w, t) \geq 0.$$

Proof. Absolute continuity of $B^{\leq}(\cdot, t)$ follows from $b(\cdot, t) \in L^1(\mathbb{R})$ and the fundamental theorem of calculus for Lebesgue integrals. Likewise $S^{\geq}(\cdot, t)$ is absolutely continuous since it is an integral of $a(\cdot, t) \in L^1(\mathbb{R})$ over a moving upper limit. Continuity is a consequence of absolute continuity. For a.e. w ,

$$\partial_w B^{\leq}(w, t) = b(w, t), \quad \partial_w S^{\geq}(w, t) = -a(w, t),$$

hence $\partial_w D = b + a = \rho \geq 0$ a.e., which implies D is nondecreasing. $\square$

Potential (energy). Fix a reference wage $w_0 \in \mathbb{R}$ and define

$$U(w, t) := \int_{w_0}^w D(x, t) dx.$$

Then $U(\cdot, t)$ is convex since $\partial_{ww}U(\cdot, t) = \rho(\cdot, t) \geq 0$ a.e.

Proposition 5 (Coercivity). *Under Assumption 3, $U(\cdot, t)$ is coercive: $U(w, t) \rightarrow +\infty$ as $w \rightarrow \pm\infty$.*

Proof. By dominated convergence and $b(\cdot, t) \in L^1$, we have $B^\leq(w, t) \rightarrow 0$ as $w \rightarrow -\infty$ and $B^\leq(w, t) \rightarrow B(t)$ as $w \rightarrow +\infty$. Similarly, $S^\geq(w, t) \rightarrow A(t)$ as $w \rightarrow -\infty$ and $S^\geq(w, t) \rightarrow 0$ as $w \rightarrow +\infty$. Therefore

$$\lim_{w \rightarrow -\infty} D(w, t) = -A(t) < 0, \quad \lim_{w \rightarrow +\infty} D(w, t) = B(t) > 0.$$

Hence there exist M_- and $c_- > 0$ such that for all $w \leq -M_-$, $D(w, t) \leq -c_-$; and M_+ and $c_+ > 0$ such that for all $w \geq M_+$, $D(w, t) \geq c_+$. For $w \geq M_+$,

$$U(w, t) = U(M_+, t) + \int_{M_+}^w D(x, t) dx \geq U(M_+, t) + c_+(w - M_+) \rightarrow +\infty.$$

For $w \leq -M_-$,

$$U(w, t) = U(-M_-, t) - \int_w^{-M_-} D(x, t) dx \geq U(-M_-, t) + c_-(-M_- - w) \rightarrow +\infty,$$

since $-D(x, t) \geq c_-$ on that tail. Thus $U(\cdot, t)$ is coercive. $\square$

Theorem 2 (Equilibrium as minimizer; uniqueness under thickness). *Fix t and suppose Assumption 3 holds. Then:*

1. *There exists at least one clearing wage $w^*(t) \in \mathbb{R}$ such that $D(w^*(t), t) = 0$.*
2. *Any such $w^*(t)$ is a global minimizer of $U(\cdot, t)$.*
3. *If there exists an interval I containing $w^*(t)$ such that $\rho(w, t) \geq \lambda > 0$ for a.e. $w \in I$, then the clearing wage is unique.*

Proof. (1) By Proposition 5 we have tail signs for D : $D(w, t) < 0$ for w sufficiently negative and $D(w, t) > 0$ for w sufficiently positive. Since $D(\cdot, t)$ is continuous (Lemma 4), the intermediate value theorem yields at least one zero.

(2) Since $\partial_w U(\cdot, t) = D(\cdot, t)$ and $U(\cdot, t)$ is convex, any point with $D(w, t) = 0$ is a minimizer. Coercivity implies the minimum is global.

(3) Suppose two distinct zeros $w_1 < w_2$ lie in an interval where $\rho \geq \lambda > 0$ a.e. Then

$$0 = D(w_2, t) - D(w_1, t) = \int_{w_1}^{w_2} \partial_w D(w, t) dw = \int_{w_1}^{w_2} \rho(w, t) dw \geq \lambda(w_2 - w_1) > 0,$$

a contradiction. Hence the zero is unique. $\square$

Remark 7 (Thickness/stiffness parameter). A convenient local thickness parameter is

$$\kappa(t) := \rho(w^*(t), t) = a(w^*(t), t) + b(w^*(t), t),$$

when ρ is continuous at $w^*(t)$. Larger κ means a steeper restoring slope near clearing and thus less sensitivity to perturbations.

Manipulation/perturbation bound (the thin-market channel). To make “thin markets are easier to move” precise, compare two environments with imbalances D_0 and D_1 at the same time t .

Proposition 6 (Sensitivity of clearing wage to bounded deformations). *Fix t and suppose $D_0(\cdot, t)$ is continuous and strictly increasing on an interval containing its unique zero $w_0^*(t)$, with $\rho_0(w, t) \geq \lambda > 0$ a.e. on that interval. Let $D_1(\cdot, t)$ be another continuous imbalance function and define the uniform deformation size*

$$\varepsilon := \sup_{w \in \mathbb{R}} |D_1(w, t) - D_0(w, t)|.$$

If $D_1(\cdot, t)$ has a (necessarily unique) zero $w_1^(t)$, then*

$$|w_1^*(t) - w_0^*(t)| \leq \frac{\varepsilon}{\lambda}.$$

Proof. By strict monotonicity of D_0 and $\rho_0 \geq \lambda$ a.e., for any w in the interval we have the slope bound

$$D_0(w, t) - D_0(w_0^*, t) = \int_{w_0^*}^w \rho_0(x, t) dx \geq \lambda(w - w_0^*).$$

Since $D_0(w_0^*, t) = 0$, this implies $D_0(w, t) \geq \lambda(w - w_0^*)$ for $w \geq w_0^*$ and $D_0(w, t) \leq \lambda(w - w_0^*)$ for $w \leq w_0^*$.

Now evaluate at $w = w_1^*$ where $D_1(w_1^*, t) = 0$:

$$|D_0(w_1^*, t)| = |D_0(w_1^*, t) - D_1(w_1^*, t)| \leq \varepsilon.$$

If $w_1^* \geq w_0^*$, then $D_0(w_1^*, t) \geq \lambda(w_1^* - w_0^*)$, so $\lambda(w_1^* - w_0^*) \leq \varepsilon$. If $w_1^* \leq w_0^*$, then $-D_0(w_1^*, t) \geq \lambda(w_0^* - w_1^*)$, so $\lambda(w_0^* - w_1^*) \leq \varepsilon$. Combine both cases to get the bound. $\square$

Remark 8 (Interpretation). The bound scales like $1/\lambda$: when local thickness λ is small, a fixed deformation size ε produces a large wage shift. This is the precise “thin markets are manipulable” mechanism used later in the predation criteria.

5.5 Trust-mediated participation and endogenous thickness

The clearing objects above admit a population interpretation on the same measurable worker space used by the finite-horizon selection problem. Let (I, Σ, μ) be a worker space, let $\delta : I \rightarrow (0, 1)$ be the patience field, and let $c_t : I \rightarrow \mathbb{R}_+$ be the date- t cost of maintaining trusted repeated participation. For a protocol P relative to an outside option S , worker i evaluates

$$\Delta U(i) := \sum_{s=1}^T \delta(i)^{s-1} \Delta_s(i),$$

so Theorem 1 applies pointwise and induces the measurable acceptance set

$$A_P := \{i \in I : \Delta U(i) \geq 0\}.$$

Aggregate selection enters wage clearing through the reservation-wage mass generated by these worker-indexed objects.

Repeated labor relations are encoded by a measurable kernel $K_t(dj \mid i)$ on I and a relational-quality map $q_t : I \times I \rightarrow [0, 1]$. For any measurable participation field $p_t : I \rightarrow [0, 1]$,

define

$$(\mathcal{Q}_t p_t)(i) := \int_I p_t(j) q_t(i, j) K_t(dj \mid i).$$

Let $s_t : I \rightarrow \mathbb{R}_+$ measure the surplus carried by a reliable repeated relation. The trusted continuation-value field is

$$V_t^\tau(i) := \frac{s_t(i)}{1 - \delta(i)} (\mathcal{Q}_t p_t)(i).$$

A convenient measurable participation rule is

$$p_t(i) := \mathbf{1}_{\{V_t^\tau(i) \geq c_t(i)\}}.$$

This rule induces a fixed-point problem on measurable fields: patience raises the continuation multiplier, the relation kernel maps participation into relational quality, and the resulting continuation value determines the next participation field.

For the ask side, let $\Phi_t(\cdot \mid i)$ be worker i 's reservation-wage measure at date t . The trust-mediated ask measure is

$$A_t^\tau(E) := \int_I p_t(i) \Phi_t(E \mid i) \mu(di), \quad E \in \mathcal{B}(\mathbb{R}).$$

If $\Phi_t(\cdot \mid i)$ admits a density $\varphi(w \mid i, t)$, then

$$a^\tau(w, t) = \int_I p_t(i) \varphi(w \mid i, t) \mu(di).$$

Writing the total ask density as

$$a(w, t) = a^u(w, t) + a^\tau(w, t),$$

the local stiffness parameter becomes

$$\kappa(t) = a^u(w^*(t), t) + a^\tau(w^*(t), t) + b(w^*(t), t).$$

Proposition 7 (Trust-mediated participation raises local thickness). *Fix t and consider two environments $e \in \{0, 1\}$ that share the same kernel K_t , relational-quality map q_t , reservation-wage density $\varphi(\cdot \mid i, t)$, untrusted ask density $a^u(\cdot, t)$, and bid density $b(\cdot, t)$. Define*

$$a_e^\tau(w, t) := \int_I p_{e,t}(i) \varphi(w \mid i, t) \mu(di), \quad \rho_e(w, t) := a^u(w, t) + a_e^\tau(w, t) + b(w, t).$$

If $p_{1,t}(i) \geq p_{0,t}(i)$ for μ -a.e. i , then

$$a_1^\tau(w, t) \geq a_0^\tau(w, t) \quad \text{for all } w \in \mathbb{R},$$

and hence

$$\rho_1(w, t) \geq \rho_0(w, t) \quad \text{for all } w \in \mathbb{R}.$$

Consequently every local thickness lower bound available in environment 0 is weakly increased in environment 1. In particular, if $\rho_0(w, t) \geq \lambda_0 > 0$ on an interval J containing $w_0^*(t)$, then environment 1 admits a lower bound $\rho_1(w, t) \geq \lambda_1 \geq \lambda_0$ on J , and the perturbation estimate in Proposition 6 contracts from ε/λ_0 to ε/λ_1 .

Proof. The density inequality follows immediately from nonnegativity of $\varphi(\cdot \mid i, t)$ and the

assumption $p_{1,t} \geq p_{0,t}$ μ -a.e.:

$$a_1^\tau(w, t) - a_0^\tau(w, t) = \int_I (p_{1,t}(i) - p_{0,t}(i)) \varphi(w | i, t) \mu(di) \geq 0.$$

Adding the common components a^u and b yields $\rho_1 \geq \rho_0$ pointwise. The statement about lower bounds on J follows directly, and Proposition 6 then yields the tighter perturbation estimate. $\square$

A patience shift that expands trusted participation therefore increases the mass of reservation wages carried into the clearing problem. Under the measurable rule above, the dynamic channel is

$$\delta \uparrow \implies p_t \uparrow \implies a^\tau(\cdot, t) \uparrow \implies \kappa(t) \uparrow,$$

with the last step evaluated at the clearing wage. The same worker distribution thus governs both acceptance of extractive protocols and the stiffness of wage clearing.

5.6 Dynamics: gradient flow, finite temperature, and finite- n noise

Deterministic adjustment as gradient flow. Assume for this local adjustment analysis that macro time t is fixed, so wage dynamics evolve under the contemporaneous potential $U(\cdot, t)$. Later on we shall allow the clearing environment to vary smoothly over t . A minimal adjustment rule is

$$\dot{w}(\tau) = -\frac{1}{\gamma} \partial_w U(w(\tau), t) = -\frac{1}{\gamma} D(w(\tau), t), \quad \gamma > 0, \quad (8)$$

where τ is the adjustment time-scale.²

Assumption 4 (Local Lipschitz for well-posedness). For each fixed t , the map $w \mapsto D(w, t)$ is locally Lipschitz on $\mathbb{R}$. (A sufficient condition is that $a(\cdot, t)$ and $b(\cdot, t)$ are locally bounded and locally of bounded variation; in many applications, continuity plus local boundedness is already enough after mollification.)

Theorem 3 (Existence/uniqueness and Lyapunov descent). *Fix t and suppose Assumptions 3 and 4 hold. Then:*

1. *For any initial wage $w(0) = w_0$, the ODE (8) has a unique maximal solution $w(\tau)$, and the solution is global (i.e. exists for all $\tau \geq 0$).*
2. *Along any solution, $U(w(\tau), t)$ is nonincreasing and*

$$\frac{d}{d\tau} U(w(\tau), t) = -\frac{1}{\gamma} D(w(\tau), t)^2 \leq 0.$$

3. *If $D(\cdot, t)$ has a unique zero $w^*(t)$ and is strictly increasing in a neighborhood of $w^*(t)$, then $w(\tau) \rightarrow w^*(t)$ as $\tau \rightarrow \infty$.*

Proof. (1) Local existence and uniqueness follow from Picard–Lindelöf since $D(\cdot, t)$ is locally Lipschitz. To show global existence, it suffices to rule out finite-time blow-up. By Proposition 5, $U(\cdot, t)$ is coercive. From (2) below, $U(w(\tau), t) \leq U(w_0, t)$ for all τ in the

²We keep t as “macro time” and τ as within- t adjustment. One can set $\tau = t$ if desired.

interval of existence; thus $w(\tau)$ remains in the compact sublevel set $\{w : U(w, t) \leq U(w_0, t)\}$, preventing blow-up and extending the solution globally.

(2) Differentiate $U(w(\tau), t)$ in τ :

$$\frac{d}{d\tau}U(w(\tau), t) = \partial_w U(w(\tau), t) \dot{w}(\tau) = D(w(\tau), t) \cdot \left(-\frac{1}{\gamma} D(w(\tau), t) \right) = -\frac{1}{\gamma} D(w(\tau), t)^2 \leq 0.$$

(3) Since $U(\cdot, t)$ is convex and coercive, it has at least one minimizer; by the unique-zero assumption, the minimizer is unique and equals $w^*(t)$. The Lyapunov identity implies $D(w(\tau), t) \in L^2([0, \infty))$, and the trajectory stays in a compact set. Any limit point $\bar{w}$ of $w(\tau)$ must satisfy $D(\bar{w}, t) = 0$; otherwise, by continuity of D , D^2 would stay bounded away from zero on a neighborhood, contradicting $\int_0^\infty D(w(\tau), t)^2 d\tau < \infty$. Thus every limit point equals $w^*(t)$, implying convergence $w(\tau) \rightarrow w^*(t)$. $\square$

Exponential rate under strong thickness. If $\rho(w, t) \geq \lambda > 0$ a.e. on an interval containing $w^*(t)$, then $D(\cdot, t)$ is strongly monotone there and the convergence becomes exponential once $w(\tau)$ enters that neighborhood:

$$|w(\tau) - w^*(t)| \leq e^{-(\lambda/\gamma)(\tau-\tau_0)} |w(\tau_0) - w^*(t)|.$$

(A short proof follows by differentiating $|w - w^*|^2$ and using $D(w) \geq \lambda(w - w^*)$ as in Proposition 6.)

Moving equilibrium and a tracking identity. Now allow $t \mapsto D(\cdot, t)$ to vary smoothly.

Assumption 5 (Smoothness in time near clearing). There exists a neighborhood of the graph $\{(w^*(t), t)\}$ on which $(w, t) \mapsto D(w, t)$ is C^1 and $\partial_w D(w^*(t), t) = \kappa(t) > 0$.

Proposition 8 (Speed of the moving clearing wage). *Under Assumption 5, the clearing wage path $t \mapsto w^*(t)$ is C^1 and satisfies*

$$\dot{w}^*(t) = -\frac{\partial_t D(w^*(t), t)}{\partial_w D(w^*(t), t)} = -\frac{\partial_t D(w^*(t), t)}{\kappa(t)}. \quad (9)$$

Proof. Apply the implicit function theorem to the equation $D(w, t) = 0$ at $(w^*(t), t)$ using $\partial_w D(w^*(t), t) = \kappa(t) > 0$. Differentiating $D(w^*(t), t) = 0$ in t yields (9). $\square$

This identity makes stiffness explicit: for a fixed disturbance $\partial_t D$, thicker markets (large κ) move less.

Finite temperature / finite- n noise as Langevin dynamics. To model finite participation (finite n agents, noisy offers) we add stochastic forcing:

$$dw_\tau = -\frac{1}{\gamma} \partial_w U(w_\tau, t) d\tau + \sqrt{\frac{2T}{\gamma}} dW_\tau, \quad T > 0. \quad (10)$$

This is the overdamped Langevin (Smoluchowski) diffusion in the potential $U(\cdot, t)$.

Assumption 6 (Smooth coercive potential). Fix t . Assume $a(\cdot, t)$ and $b(\cdot, t)$ are C^1 with integrable derivatives, so that $D(\cdot, t) \in C^1$ and $U(\cdot, t) \in C^2$, and that $U(\cdot, t)$ is coercive and $\partial_w U(\cdot, t)$ is globally Lipschitz (or at least of at most linear growth).

Proposition 9 (Well-posedness and invariant density (time-homogeneous case)). *Assume 6 and that $U(\cdot, t)$ is time-homogeneous during adjustment (fix t). Then (10) has a unique strong solution for all $\tau \geq 0$. Moreover, the density*

$$\varrho_\infty(w) = \frac{1}{Z} \exp\left(-\frac{U(w, t)}{T}\right), \quad Z := \int_{\mathbb{R}} \exp\left(-\frac{U(w, t)}{T}\right) dw < \infty,$$

is invariant for the diffusion.

Proof. Existence/uniqueness of a strong solution follows from standard SDE theory under global Lipschitz (or linear growth + local Lipschitz) of the drift. Coercivity ensures $Z < \infty$ because $U(w, t) \rightarrow \infty$ implies $\exp(-U/T)$ is integrable.

To verify invariance, write the Fokker–Planck equation for the density $\varrho(w, \tau)$:

$$\partial_\tau \varrho = \frac{1}{\gamma} \partial_w (\varrho \partial_w U + T \partial_w \varrho).$$

Plugging $\varrho_\infty \propto e^{-U/T}$ gives $\partial_w \varrho_\infty = -(1/T) \varrho_\infty \partial_w U$, hence

$$\varrho_\infty \partial_w U + T \partial_w \varrho_\infty = \varrho_\infty \partial_w U - T \cdot \frac{1}{T} \varrho_\infty \partial_w U = 0,$$

so the probability flux vanishes and $\partial_\tau \varrho_\infty = 0$. $\square$

Finite- n microfoundation for the noise scale (consistent with D). One concrete “finite n ” model is: at adjustment step τ there are n_B independent firm offers (bids) W_i^B drawn from the *normalized* density $\tilde{b}(\cdot, t) := b(\cdot, t)/B(t)$, and n_A independent worker reservation wages (asks) W_j^A drawn from $\tilde{a}(\cdot, t) := a(\cdot, t)/A(t)$. Define scaled empirical cumulatives

$$\hat{B}_n^\leq(w, t) := \frac{B(t)}{n_B} \sum_{i=1}^{n_B} \mathbf{1}\{W_i^B \leq w\}, \quad \hat{S}_n^\geq(w, t) := \frac{A(t)}{n_A} \sum_{j=1}^{n_A} \mathbf{1}\{W_j^A \geq w\},$$

and the empirical imbalance

$$\hat{D}_n(w, t) := \hat{B}_n^\leq(w, t) - \hat{S}_n^\geq(w, t).$$

Then $\mathbb{E}[\hat{B}_n^\leq(w, t)] = B^\leq(w, t)$ and $\mathbb{E}[\hat{S}_n^\geq(w, t)] = S^\geq(w, t)$, hence

$$\mathbb{E}[\hat{D}_n(w, t)] = D(w, t).$$

Moreover,

$$\text{Var}(\hat{B}_n^\leq(w, t)) = \frac{B(t)^2}{n_B} p_B(w, t)(1 - p_B(w, t)), \quad p_B(w, t) := \frac{B^\leq(w, t)}{B(t)},$$

and similarly

$$\text{Var}(\hat{S}_n^\geq(w, t)) = \frac{A(t)^2}{n_A} p_A(w, t)(1 - p_A(w, t)), \quad p_A(w, t) := \frac{S^\geq(w, t)}{A(t)}.$$

Thus the imbalance noise $\xi_n(w, t) := \hat{D}_n(w, t) - D(w, t)$ has variance of order $O(1/n_B + 1/n_A)$. If wage adjustment uses $\hat{D}_n$ in place of D , the update becomes a noisy gradient step whose noise amplitude is $O(n^{-1/2})$ for an effective sample size n (e.g. $n := \min\{n_A, n_B\}$). Under a small step-size / diffusion approximation, this yields an overdamped Langevin limit of

the form (10), with “temperature” T proportional to the local variance of the empirical imbalance.

5.7 From “surplus” to “extraction”: when does surplus capture become predatory?

We use “exploitation” in the Marxian technical sense only when *surplus capture is structurally enforced and not primarily compensation for risk, coordination, or genuine entrepreneurship*. Within the present dynamic primitives, three sufficient (not necessary) conditions emerge.

(A) Thin markets / low stiffness enable manipulation rents. If $\kappa(t) = \rho(w^*(t), t)$ is small, then 9 implies that a given deformation of the bid/ask environment produces a large movement in clearing wages. When small administrative or collusive interventions can cheaply shift $D(\cdot, t)$, rents can be extracted without commensurate productive contribution.

(B) High time preference selects workers into dominated contracts. If outside-option increments exhibit the single-crossing profile of 3, then 1 implies sufficiently impatient workers accept contracts/institutions that are negative in total. This is a formal selection channel for “extraction by horizon engineering.”

(C) Coercive outside option + bounded mobility turns surplus into structural extraction. Even if an individual contract is formally voluntary, the system becomes Marx-exploitative when the feasible exit set is small enough that extractive protocols are effectively forced in equilibrium: the environment shifts Δ_t so that $\delta^* \rightarrow 1$ (almost everyone accepts), or exit is blocked by institutional constraints. In this case, surplus capture is not an occasional rent but an equilibrium feature of reproduction.

6 Discussion: when does surplus capture from labor become Marxian exploitation?

The preceding sections separate three ideas that are often conflated in popular debates: (i) *profits* as an accounting residual, (ii) *extraction* as durable surplus capture enabled by power or barriers, and (iii) *Marxian exploitation* as a structurally enforced class relation in which wage-labor produces a surplus that is appropriated by owners/controllers through control rights and property. This distinction matters because profits can be large even when the labor wedge is zero (Proposition 1), and because many modern surplus channels operate through rents and transfers rather than through the wage relation.

6.1 A checklist: necessary structure and contingent amplifiers

A practical way to state the paper’s claim is as a conjunction: *Marxian exploitation requires a core structure*, while its *intensity* (depicted here as “predatory and dominant” surplus capture from labor) depends on contingent wedge-enablers.

Core structure (necessary). Surplus capture is Marxian exploitation only if:

1. labor is purchased as labor-power for a wage;
2. production is controlled by owners/controllers who appropriate the residual;

3. workers lack independent access to the means of production (compulsion by property).

These are the structural conditions under which surplus labor is appropriated as surplus value [7].

Contingent wedge-enablers (sufficient for intensity/predation). Exploitation becomes *predatory and dominant*—i.e. surplus capture *from labor* is amplified beyond competition-compatible returns—when, in addition:

1. **Exit is constrained:** monopsony, search frictions, or mobility limits keep w below $MRPL$ [6, 9].
2. **Enforcement is cheap:** monitoring technologies and organizational layers reduce the cost of intensifying labor and sustaining wedges [1, 24].
3. **Rents raise the prize:** markups, licensing, or other barriers increase the return to investing in control [3].
4. **Institutions are “closed access”:** bureaucratic allocation and weak accountability sustain rents and suppress voice [10, 11].
5. **Short horizons dominate:** worker impatience sustains acceptance of front-loaded/back-loaded-negative contracts, and producer impatience sustains wage refusal and under-investment in retention/skill.
6. **Low trust thins markets:** segmentation lowers the “stiffness” of wage clearing, making suppression interventions more powerful [15].
7. **Governance rewards extraction:** financialized payout regimes create systematic pressure to convert control into wedge profits [5].

The list does not discard Marx’s core insight; it clarifies what is *structural* (wage-labor, control, residual claimancy) and what is *contingent* (the size and enforceability of wedges, and the institutional apparatus that stabilizes them). In this view, Marxian exploitation is always *possible* under capitalism; whether it becomes *dominant and predatory* is an equilibrium outcome of slow-moving parameters governing outside options, enforcement costs, rents, trust, and horizons.

6.2 How dynamics play into protocol selection and market stiffness

The dynamic primitives in Section 5 provide two complementary selection mechanisms that connect the checklist to persistence.

Horizon engineering selects agents into dominated arrangements. The finite-horizon cutoff theorem (Theorem 1) formalizes a ubiquitous pattern of extractive labor relations: *front-loaded sweeteners with back-loaded burn*. On a worker space (I, Σ, μ) , the theorem acts pointwise through the field $\Delta U(i) = \sum_{t=1}^T \delta(i)^{t-1} \Delta_t(i)$ and induces the acceptance set

$$A_P := \{i \in I : \Delta U(i) \geq 0\}.$$

When increments exhibit a single sign change (positive early, negative later), there exists a unique cutoff δ^* such that sufficiently impatient workers accept the protocol even when its undiscounted sum is negative. This turns “short horizons” into a measurable selection

channel: environments that raise effective time preference (liquidity constraints, debt, unstable housing, or high background risk) expand the mass $\mu(A_P)$.

The cognition channel in Section 5.2 sharpens the same logic under risk: if subjective probability weighting distorts the evaluation of tail losses or lottery-like upside, the relevant horizon profile becomes cognition-dependent, shifting the acceptance cutoff to $\delta^*(g)$. Lower cognition (more distortion) raises the population mass that accepts protocols whose fully informed long-run evaluation is negative.

Trust-mediated thickness governs the wage-clearing channel. Section 5.5 lifts wage clearing to the same measurable worker space. Trusted repeated relations generate a participation field p_t , that field generates a trust-mediated ask measure A_t^T , and the induced density $a^\tau(w, t)$ enters the local stiffness parameter $\kappa(t)$. A rise in patience increases the continuation value attached to trusted relations, expands trusted participation under a monotone measurable rule, raises reservation-wage mass near clearing, and tightens the perturbation bound in Proposition 6. Segmentation, gatekeeping, and administrative chokepoints work through the same mechanical channel by keeping participation thin near $w^*(t)$.

Taken together, the dynamic objects map the joint distribution of patience and cognition into two measurable outcomes: the acceptance mass for extractive protocols and the stiffness of wage clearing. Persistence follows when the acceptance mass is large and the local thickness lower bound is small. High-trust, long-horizon environments compress both margins and shrink the feasible wedge set.

6.3 Two practical implications

Measurement. Measured profits or capital share alone cannot identify Marxian exploitation. One needs evidence that surplus capture is generated *from labor* via wedges (mobility constraints, monopsony, suppressed outside options, enforcement apparatus), rather than primarily from rents or transfers (Proposition 1). In practice this means focusing on institutions and microstructure: the elasticity of labor supply to the firm, credible outside options, enforceability of exit, and the technologies and layers that sustain wage suppression.

Comparative institutional prediction. High-trust, long-horizon, open-access orders can sustain lower predatory exploitation without heavy direct control: thick participation raises stiffness and patient agents reject extractive intertemporal profiles, reducing the feasible wedge set. Conversely, closed-access bureaucratic orders and financialized governance jointly reduce exit, cheapen enforcement, and increase the return to control, making exploitation a stable outcome of reproduction rather than an episodic abuse. The paper’s dynamic language makes the comparative prediction crisp: extraction persists where (i) many workers satisfy $\delta \leq \delta^*$ (or $\delta \leq \delta^*(g)$ under distorted risk), and (ii) wage clearing is mechanically easy to shift (low κ) via small institutional deformations.

6.4 Implications for Marxian structural analysis

This framework is compatible with Marx’s structural insight while refusing an over-strong empirical inference from profits to exploitation. It does *not* claim that rents are normatively benign, nor that modern capitalism reduces to one wedge. It claims that calling surplus capture “Marxian exploitation” is a statement about a specific relation—wage-labor under control and compulsion by property—and that the degree to which that relation becomes predatory depends on identifiable, interacting mechanisms: labor-market power, enforcement

technologies, rent opportunities, institutional closure, governance horizons, trust, and cognition.

Marxian exploitation is a structural relation rooted in wage-labor and control of production, but whether surplus capture *from labor* becomes predatory depends on a large set of contingent factors. A modern synthesis separates the *concept* (surplus labor appropriated by controllers) from the *channels* that intensify it (labor-market power, rent opportunities, state distortion, organizational enforcement, financial governance, and the joint distribution of time preference, cognition, and trust). Modeling these channels dynamically under finite horizons yields a unified prediction: extractive labor relations persist when impatient or distortedly optimistic workers accept front-loaded/back-loaded-negative protocols and impatient producers refuse wage investment, especially in thin, manipulable markets; conversely, high-trust, long-horizon environments thicken participation and collapse the feasible wedge space, reducing predatory exploitation even in the absence of pervasive direct state control.

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